

Kinetic theory of stimulated Brillouin scattering with velocity-dependent collisional relaxation

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In this work, we develop a kinetic theory of stimulated Brillouin scattering (SBS) based on a generalized Bhatnagar-Gross-Krook (BGK) model incorporating a velocity-dependent collision frequency. Starting from the Boltzmann equation, we derive an analytic expression for the SBS gain coefficient that explicitly captures the coupling between velocity-dependent relaxation and acoustic response. We show that this modification leads to a systematic reshaping of the gain spectrum, including broadening, peak enhancement, and pressure-dependent frequency shifts that are not captured by conventional BGK theory. Our results demonstrate improved agreement with experimentally measured gain coefficients, identifying velocity-resolved relaxation as a key physical mechanism underlying SBS in rarefied and intermediate collisional regimes. This framework provides a physically grounded extension of kinetic SBS theory and introduces a parameter for controlling and optimizing Brillouin gain in high-power laser systems.

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I. INTRODUCTION

Stimulated Brillouin Scattering (SBS) is a nonlinear optical process that involves the interaction of light with acoustic phonons in a medium, theoretically predicted by Brillouin [1]. Early investigations by researchers [2,3] have established the theoretical foundations, including threshold conditions, gain spectrum, and the role of medium parameters such as acoustic velocity and refractive index [4]. The first observation of SBS was reported in 1964 in quartz and sapphire crystals [5], followed by subsequent demonstrations in liquids [6–8] and gases [9]. Soon after, Kroll provided a theoretical description of the effect using a classical model based on electrostriction and photoelasticity [10]. Later, Tang extended Kroll's theory by incorporating nonlinear interactions between coupled waves, including saturation effects [11] and a few years later, the first combined experimental and theoretical study of phonon lifetime was published [12]. Over the past few decades, SBS has been extensively studied due to its significant implications in optical communications [13–17], fiber sensing [18–20], and Brillouin spectroscopy [21,22].

Theoretical descriptions of SBS in gases commonly rely on kinetic models derived from the Boltzmann transport equation. Among these approaches, the Bhatnagar-Gross-Krook (BGK) approximation provides a tractable framework by replacing the full collision integral with a relaxation term driving the distribution function toward local equilibrium [23]. The kinetic approach, particularly with a BGK collision

operator, can accurately capture the growth rate of the SBS instability, including the transition from kinetic to fluidlike behavior as the collision rate increases [24]. The Boltzmann kinetic equation together with the BGK relaxation model has also been used to describe stimulated molecular scattering in polyatomic gases across collisionless and hydrodynamic regimes [25].

Experimental measurements of SBS gain spectra reveal systematic deviations from predictions of conventional BGK theory, including discrepancies in spectral width, peak amplitude, and frequency shift [26]. These observations suggest that additional collisional effects beyond velocity-independent relaxation may influence the SBS response.

More broadly, both experimental and theoretical studies have addressed relaxation phenomena in rarefied polyatomic gases and their manifestation in Rayleigh-Brillouin scattering. Experimental investigations of Rayleigh-Brillouin scattering in molecular gases have been reported in Refs. [27–29], while theoretical descriptions of relaxation phenomena based on continuum models, generalized kinetic formulations, and rational extended thermodynamics have been developed in Refs. [30–33]. These works provide important insight into nonequilibrium relaxation processes, and collisional dynamics in gases. In the present work, we examine how a velocity-dependent collision frequency modifies the SBS gain spectrum within a generalized BGK framework that retains analytical tractability while capturing the essential collisional physics. The investigation of the BGK model incorporating a velocity-dependent collision frequency was carried out by Struchtrup [34], who showed that adopting a simple power-law dependence for the collision frequency yields a more realistic Prandtl number consistent with empirical observations. However, the influence of velocity-dependent relaxation

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on stimulated Brillouin scattering has not been systematically investigated within a consistent kinetic framework.

In this work, we develop a kinetic theory of stimulated Brillouin scattering based on a generalized BGK model with velocity-dependent relaxation time. Starting from the Boltzmann equation, we derive an analytical expression for the SBS gain coefficient that explicitly accounts for velocity-resolved collisional dynamics. We show that this extension leads to qualitative and quantitative modifications of the gain spectrum, including broadening, enhancement, and pressure-dependent shifts of the Brillouin resonance. Importantly, the present model recovers the standard BGK results in the limit of vanishing velocity dependence [25], thereby providing a systematic generalization rather than an alternative formulation. Comparison with available experimental data demonstrates that incorporating velocity-dependent relaxation significantly improves agreement with observed gain coefficients and spectral features. These results identify velocity-dependent collisional dynamics as a key mechanism governing SBS in gases and establish a framework for engineering Brillouin gain in high-power laser systems, including applications in inertial confinement fusion and gas-based pulse compression.

II. THEORETICAL MODEL

To model the microscopic dynamics of a dilute gas, we consider the Boltzmann transport equation for the distribution function $f(\vec{r}, \vec{v}, t)$, which describes the probability density of finding a particle in position $\vec{r}$ with velocity $\vec{v}$ at time t ,

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla\right)f(r, v, t) + \frac{\mathbf{F}}{m} \cdot \nabla_v f(r, v, t) = \left(\frac{\partial f}{\partial t}\right)_c, \quad (1)$$

where $\mathbf{F}$ is the external force acting on the particles, m is the particle mass, and $(\partial f/\partial t)_c$ denotes the collision integral, accounting for binary interactions between particles. To render the equation analytically tractable while retaining essential collisional dynamics, we adopt the BGK approximation [23]. In this model, the complex collision integral is replaced by a single relaxation term driving the distribution function toward a local Maxwell-Boltzmann equilibrium $(\partial f/\partial t)_c = (f_0 - f)/\tau$. Since we consider gases possessing internal degrees of freedom [35], we write the collision integral in the form

$$\left(\frac{\partial f}{\partial t}\right)_c = \frac{M_t - f}{\tau_{el}(v)} + \frac{M_i - f}{\tau_{in}(v)}, \quad (2)$$

where $M_t = n\phi(T_t)\psi_l(T_t)$ and $M_i = n\phi(T_T)\psi_l(T_T)$ with

$$\phi(T_r) = \left(\frac{m}{2\pi T_r}\right)^{3/2} \exp\left[-\frac{m(v - v_{loc})^2}{2T_r}\right],$$

$$\psi_l(T_j) = \frac{e^{-E_l/T_j}}{\sum_s e^{-E_s/T_j}}. \quad (3)$$

Here, T_t denotes the translational temperature, T_i denotes the internal temperature, T_T denotes the total temperature of the gas, and T_0 represents the equilibrium temperature. The index $r = t, T$ corresponds to the translational and total degrees of freedom appearing in the Maxwellian distribution function $\phi(T_r)$, while the index $j = i, T$ corresponds to the internal

and total temperatures appearing in the internal-state distribution function $\psi_l(T_j)$. E_l represents the rotational energy of a molecule in the internal state l . The quantity $\tau_j(v)$ denotes the relaxation time for elastic ($j = el$) and inelastic ($j = in$) collisions, approximately given by

$$\frac{1}{\tau_j(v)} = \frac{1}{\tau_{j0}} \left(1 + \beta \frac{v}{v_0}\right), \quad (4)$$

where τ_{j0} is the equilibrium relaxation time associated with elastic and inelastic collisional processes in the absence of velocity dependence. Here, β is a dimensionless free parameter characterizing the velocity dependence of the collision frequency. Introducing the small perturbation $f = f_0 + n_0 \tilde{f}$, $n = n_0(1 + \rho)$, $v_{loc} = 0 + u$, and $T_{t,i,T} = T_0 + \Delta T_{t,i,T}$, where f_0 is the equilibrium Maxwellian distribution function, n_0 and T_0 denote the equilibrium density and equilibrium temperature, respectively, while $\tilde{f}$, δn , v_{loc} , and $\Delta T_{t,i,T}$ represent small perturbations around equilibrium. Collecting only linear terms, the collision term can be written as

$$\begin{aligned} \left(\frac{\partial f}{\partial t}\right)_c = & -\sigma_0 \left(1 + \beta \frac{v}{v_0}\right) \tilde{f} + \phi(T_0) \psi_l(T_0) \\ & \times \left(1 + \beta \frac{v}{v_0}\right) \left\{ \sigma_0 \rho + \frac{m\sigma_0}{T_0} vu \right. \\ & + \left(\frac{mv^2}{2T_0} - \frac{3}{2}\right) \left(\frac{1}{\tau_{e0}} + \frac{1}{\tau_{i0}} \frac{C_v^tr}{C_v}\right) \frac{\Delta T_t}{T_0} \\ & + \left(\frac{E_l}{T_0} - \frac{\langle E \rangle}{T_0}\right) \left(\frac{1}{\tau_{e0}} + \frac{1}{\tau_{i0}} \frac{C_v^i}{C_v}\right) \frac{\Delta T_i}{T_0} \\ & + \frac{1}{\tau_{i0}} \left\{ \frac{E_l}{T_0} - \frac{\langle E \rangle}{T_0} \right\} \frac{C_v^tr}{C_v} \frac{\Delta T_t}{T_0} \\ & \left. + \frac{1}{\tau_{i0}} \left\{ \frac{mv^2}{2T_0} - \frac{3}{2} \right\} \frac{C_v^i}{C_v} \frac{\Delta T_i}{T_0} \right\}, \end{aligned} \quad (5)$$

where $\sigma_0 = 1/\tau_{i0} + 1/\tau_{e0}$, $v_0^2 = 2T_0/m$, and $C_v = C_v^t + C_v^i$ represents the total specific heat with $C_v^t = 3/2$ and $C_v^i = (\langle E^2 \rangle - \langle E \rangle^2)/T_0^2$. Here,

$$\langle E \rangle = \sum_l E_l \psi_l(T_0) \quad (6)$$

denotes the thermal average of the internal molecular energy levels evaluated at the equilibrium temperature T_0 . The total temperature ΔT_T is eliminated by using $\Delta T_T = (C_v^t \Delta T_t + C_v^i \Delta T_i)/C_v$.

Stimulated Brillouin Scattering (SBS) is a nonlinear effect that occurs when an intense light wave (typically a laser) interacts with acoustic phonons in a medium, leading to the generation of a frequency-shifted Stokes wave and an accompanying acoustic wave. The total field is the sum of the pump laser beam E_1 and scattered Stokes wave E_2 ($E_2 || E_1$) as shown in Fig. 1,

$$E_{\text{Total}} = E_1 e^{-ik_1 z + i\omega_1 t} + E_2 e^{ik_2 z + i\omega_2 t}. \quad (7)$$

In the presence of an electromagnetic field, each molecule experiences an average force [25],

$$\mathbf{F} = \frac{\kappa}{4} \nabla |E|^2 = \frac{i\kappa}{2} q \{ E_1 E_2^* e^{i(\Omega t - qz)} + \text{c.c.} \} \hat{z}, \quad (8)$$

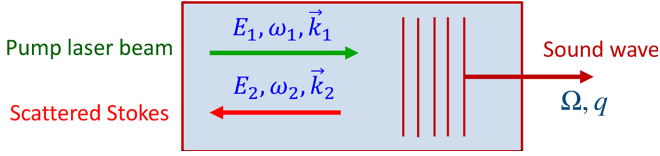


FIG. 1. Stimulated Brillouin Scattering.

where c.c. denotes the complex conjugate, $q = k_1 + k_2$, $\Omega = \omega_1 - \omega_2$, and $\kappa = (\epsilon - 1)/(4\pi n_0)$ represents the average polarizability of molecule with dielectric constant ϵ and average gas density n_0 . Introducing the slowly varying amplitudes $a = a_l e^{i(\Omega t - qz)} + \text{c.c.}$ for $\tilde{f}$, ρ , u , $\Delta T_{l,i}$, and simplifying Eq. (1), we obtain

$$f_l = \frac{A_0 \frac{\partial[f]}{\partial w} - J_l}{iqv_0(w - Z)}, \quad (9)$$

with $w = v_z/v_0$, $A_0 = -i\kappa q E_1 E_2 / (4m)$, $Z = x - iy$, $x = \Omega / (qv_0)$, $y = \sigma_0 / (qv_0)$, and $J_l = (\partial f / \partial t)_c + \sigma_0(1 + \beta v/v_0)\tilde{f}$. It is

$$\begin{pmatrix} iy(\delta + \beta R) - 1 & 0 & iu(M + \beta Q) & ir(M + \beta Q) \\ x & -1 & iu\beta(1 + i\beta y)\zeta & ir\beta(1 + i\beta y)\zeta \\ \chi & 0 & iuA - \frac{3}{2} & irA \\ 0 & 0 & iVB & isB - 1 \end{pmatrix} \begin{pmatrix} \rho \\ u \\ \frac{\Delta T_l}{T_0} \\ \frac{\Delta T_i}{T_0} \end{pmatrix} = \begin{pmatrix} -\frac{2iA_0 n_0}{qv_0^2} \left(\frac{1+ZW}{1+i\beta y} \right) \\ 0 \\ -\frac{iA_0 n_0}{qv_0^2(1+i\beta y)} \\ 0 \end{pmatrix}. \quad (11)$$

Solving Eq. (11) for the density amplitude, we obtain

$$\rho = -i \frac{2A_0}{qv_0^2} \frac{\text{Num}}{\text{Den}}, \quad (12)$$

where

$$\begin{aligned} \text{Num} &= \left\{ (1 + ZW) \left(\frac{y^2 AB}{1 + \alpha} + i(Au + 3sB/2) - 3/2 \right) \right. \\ &\quad \left. - \frac{1}{2} \left(\frac{y^2 B}{1 + \alpha} + iu \right) (M + \beta Q) \right\} / (1 + i\beta y), \\ \text{Den} &= \left(\frac{y^2 AB}{1 + \alpha} + i(Au + 3sB/2) - 3/2 \right) \\ &\quad \times (i\beta y R + iy\delta - 1) - \chi(M + \beta Q) \left(\frac{y^2 B}{1 + \alpha} + iu \right), \end{aligned}$$

with

$$\begin{aligned} A &= W + \beta(1 + i\beta y)\zeta\eta + \beta\phi, \quad \delta = W + \frac{2(1 + ZW)x}{(1 + i\beta y)}, \\ B &= W + \beta \frac{(1 + ZW)}{(1 + i\beta y)}, \quad \phi = \frac{3}{2(1 + i\beta y)} + \frac{ZW}{(1 + i\beta y)}, \\ M &= -\frac{W}{2} + \frac{Z(1 + ZW)}{(1 + i\beta y)^2}, \quad \zeta = \frac{W}{2} + \frac{Z(1 + ZW)}{(1 + i\beta y)^2}, \\ N &= -\frac{ZW}{2(1 + i\beta y)} + \frac{Z^2(1 + ZW)}{(1 + i\beta y)^3}, \\ \eta &= \frac{iy}{(1 + i\beta y)} + \frac{i\beta y Z}{(1 + i\beta y)^2} + 2i\beta y \left(\frac{W}{2} + \frac{Z}{(1 + i\beta y)^2} \right), \end{aligned}$$

important to note that the moments (local density, velocity, and temperature) related to the distribution function are given by

$$\begin{aligned} \rho &= \sum_l \int f_l d^3 v, \\ u &= \sum_l \int v f_l d^3 v, \\ \frac{\Delta T_l}{T_0} &= \sum_l \int \frac{1}{C_v^l} \left(\frac{v^2}{v_0^2} - \frac{3}{2} \right) f_l d^3 v, \\ \frac{\Delta T_i}{T_0} &= \sum_l \int \frac{1}{C_v^l} \left\{ \frac{E_l}{T_0} - \frac{\langle E \rangle}{T_0} \right\} f_l d^3 v. \end{aligned} \quad (10)$$

Substituting Eq. (9) into Eq. (10), we obtain a set of equations for all amplitudes expressed in the following matrix form,

$$\begin{aligned} \chi &= \frac{Z^2}{(1 + i\beta y)^2} + \frac{i\beta y}{2(1 + i\beta y)} - \frac{1}{2} + \frac{iyZ}{(1 + i\beta y)^2} + \eta x, \\ R &= \frac{(1 + ZW)}{(1 + i\beta y)} + 2 \frac{Z(1 + ZW)}{(1 + i\beta y)^2}, \\ Q &= N + 2iy\zeta(1 + ZW) + 2i\beta y\zeta \frac{Z(1 + ZW)}{(1 + i\beta y)}, \\ W &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} dt \frac{e^{-t^2}}{t - Z_1}, \quad Z_1 = Z - i\beta v y/v_0, \\ g &= \frac{y}{1 + \alpha} \left(1 + \frac{\alpha}{1 + \Delta} \right), \quad \alpha = \tau_{e0}/\tau_{i0}, \\ s &= \frac{y}{1 + \alpha} \left(1 + \frac{\alpha\Delta}{1 + \Delta} \right), \quad \Delta = C_v^i/C_v^e. \end{aligned}$$

Here, W is related to the error function with complex argument [25]. Starting from Maxwell's equations, we arrive at the wave equation

$$\nabla^2 E - \frac{1}{(c/n_r)^2} \frac{\partial^2 E}{\partial t^2} = \frac{4\pi}{c^2} \frac{\partial^2 \mathcal{P}}{\partial t^2}, \quad (13)$$

where nonlinear polarization $\mathcal{P} = \kappa \Delta n E$ is modulated due to change in the gas density $\Delta n = n_0 \rho$, which alters the permittivity $\Delta\epsilon = 4\pi\kappa\Delta n$. After applying a slowly varying amplitude approximation ($|\partial^2 E / \partial z^2| \ll k|\partial E / \partial z|$ and $|\partial^2 E / \partial t^2| \ll \omega|\partial E / \partial t|$) and retaining only first-order derivatives in z and t , we find

$$\frac{\partial E_2}{\partial z} - \frac{n_r}{c} \frac{\partial E_2}{\partial t} = \frac{2\pi i \omega_1}{c n_r} \mathcal{P}_2, \quad (14)$$

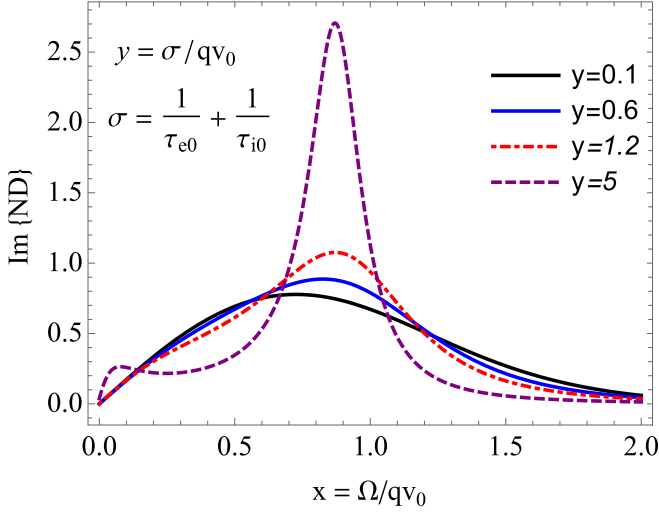


FIG. 2. Variation of the SBS gain profile $\text{Im}\{\text{ND}\}$ of the backward-scattered wave as a function of the normalized frequency $x = \Omega/(qv_0)$ for a diatomic nitrogen gas with $\alpha = \tau_{e0}/\tau_{i0} = 1/5$ and $\Delta = C_v^i/C_v^e = 2/3$, where α denotes the ratio of elastic to inelastic relaxation times and Δ represents the ratio of internal to translational heat capacities.

where

$$\mathcal{P}_2 = \frac{(n_r^2 \epsilon_0 - 1)^2}{32\pi^2 \rho_0 v_0^2} |E_1|^2 E_2(\text{ND})^*. \quad (15)$$

Here, $\text{ND} = \text{Num}/\text{Den}$. Assuming a steady-state solution ($\partial/\partial t = 0$), the stimulated Brillouin scattering gain of the backward-scattered wave is determined from Eq. (14),

$$\text{Gain} = \frac{k_1}{v_0^2 \rho_0} \frac{(n_r^2 \epsilon_0 - 1)^2}{8\pi \sqrt{\epsilon_0}} |E_1|^2 \text{Im ND}. \quad (16)$$

It is pertinent to mention here that the frequency dependence of the gain is given by the parameter ND . Figure 2 shows the imaginary part of the gain as a function of a normalized frequency $x = \Omega/(qv_0)$, for various values of dimensionless collision parameter y . This behavior highlights the critical role of relaxation processes in tuning the spectral profile and peak efficiency of stimulated Brillouin scattering in a dense media. It is pertinent to mention here that the results of Ref. [25] for stimulated Brillouin scattering gain can be recovered by substituting the velocity dependence factor $\beta = 0$. Adding velocity dependence with positive β decreases relaxation time and enhances the SBS gain (see Fig. 3).

The collision cross section is a fundamental quantity in the kinetic theory of gases and plays a critical role in determining relaxation time and scattering rate. The collision relaxation time can be written as $1/\tau_{e0(i0)} = n\sigma_{\text{el(iel)}}v_{\text{rel}}$, where $v_{\text{rel}} = \sqrt{16k_B T/\pi m}$ is the mean relative velocity between two molecules and elastic collision cross section $\sigma_{\text{el}} = 3.22 \times 10^{-19} \text{ m}^2$ [36], whereas the cross section for the inelastic collisions $\sigma_{\text{iel}} = 0.338 \times 10^{-20} \text{ m}^2$ has been found in Ref. [37]. The density of molecules follows the ideal gas law $n = P/(k_B T)$, where P denotes the pressure.

Figure 4 depicts the variation of the SBS gain with acoustic frequency $\nu = \Omega/2\pi$ in N_2 gas at pressure ranging from 1 to 6 atm. As pressure increases, the gain peak

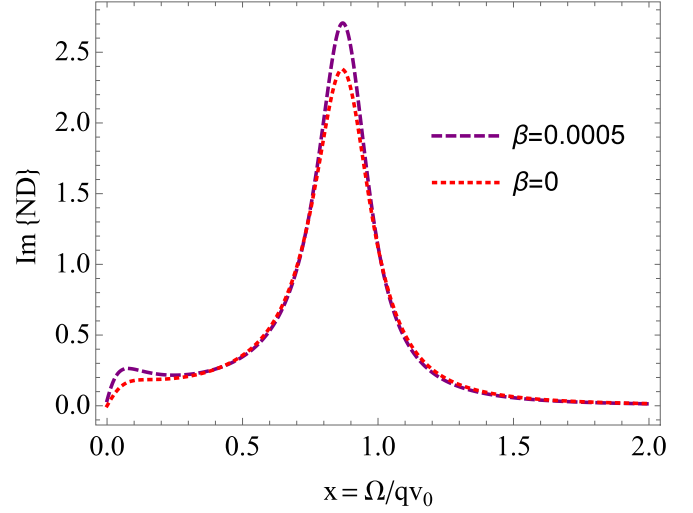


FIG. 3. Variation of SBS gain profile $\text{Im}\{\text{ND}\}$ with and without velocity dependence of the relaxation time.

shifts to higher frequencies and becomes more pronounced. Although focused on water, the authors reported the variation of SBS gain coefficient with pressure [38], which supports the general understanding that the SBS gain can be modulated through pressure adjustments in various media, including gases such as nitrogen.

Figure 5 illustrates the influence of the velocity-dependent relaxation parameter β on the SBS gain profile. The results show that velocity-dependent relaxation produces changes in both the linewidth and peak amplitude of the gain profile compared with the conventional BGK model with velocity-independent relaxation time. Although the quantitative variation remains relatively small for the parameter range considered here, the inclusion of velocity dependence introduces an additional kinetic correction to the SBS gain. It is important to compare the present theoretical results with the experimental measurements reported in Ref. [26], where deviations from predictions based on

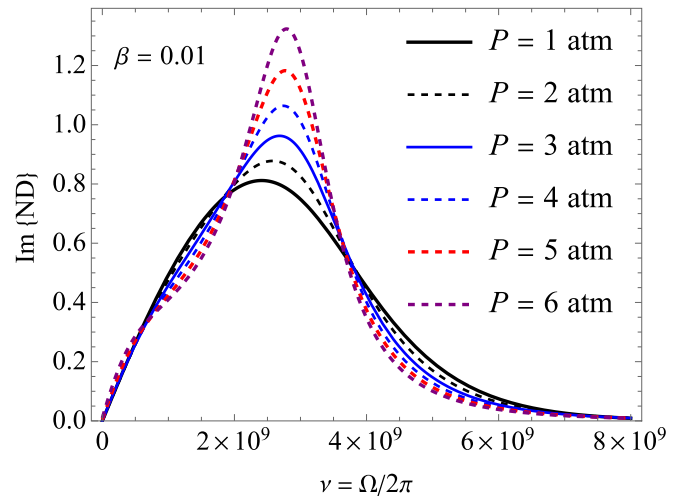


FIG. 4. Variation of SBS gain coefficient $\text{Im}\{\text{ND}\}$ as a function of frequency $\nu = \Omega/2\pi$ for nitrogen gas at different pressure $P = 1\text{--}6$ atm for fixed $\beta = 0.01$.

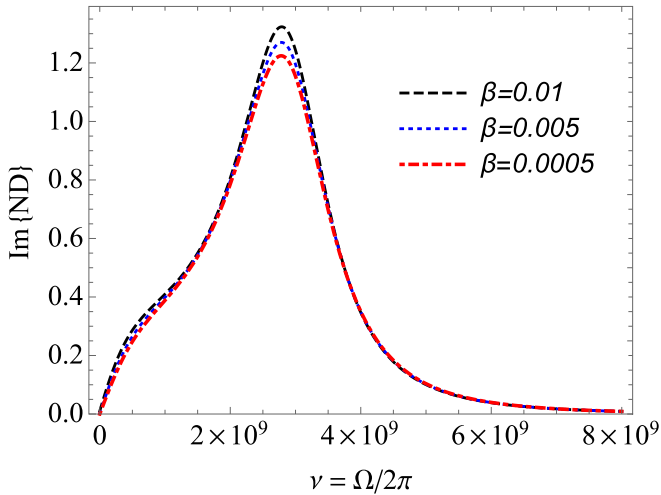


FIG. 5. Variation of SBS gain coefficient $\text{Im}\{\text{ND}\}$ as a function of frequency $\nu = \Omega/2\pi$ for nitrogen gas at different values of β for fixed pressure $P = 6$ atm.

the conventional velocity-independent BGK model were observed. In particular, Mironov *et al.* measured peak SBS gain coefficients of $\gamma_B = 0.22 \pm 0.02$ cm/GW for Kr and $\gamma_B = 0.092 \pm 0.001$ cm/GW for N_2 at $P = 6$ atm. Figure 6 presents a direct comparison between the experimentally measured SBS spectra and the theoretical gain profiles obtained from the conventional model ($\beta = 0$) and the present velocity-dependent model under identical pressure conditions. The inclusion of velocity-dependent relaxation produces a systematic modification of the spectral profile and leads to better agreement with the experimentally observed peak structure and frequency dependence. These results indicate that velocity-resolved collisional relaxation provides an additional kinetic correction that becomes relevant for describing SBS spectra in intermediate collisional regimes.

III. CONCLUSION

We have formulated a kinetic theory of stimulated Brillouin scattering that extends the standard BGK framework by incorporating a velocity-dependent collision frequency. This generalization introduces a direct coupling between collisional relaxation and the velocity distribution of particles, thereby modifying the acoustic response that mediates the Brillouin interaction. Our analysis shows that the gain spectrum is determined not only by the overall collision rate but also by its variation across velocity space. As a result, different velocity classes contribute unequally to the collective response, leading to modifications of the spectral profile that are not captured within velocity-independent models. In the appropriate limit, the present formulation reduces to the

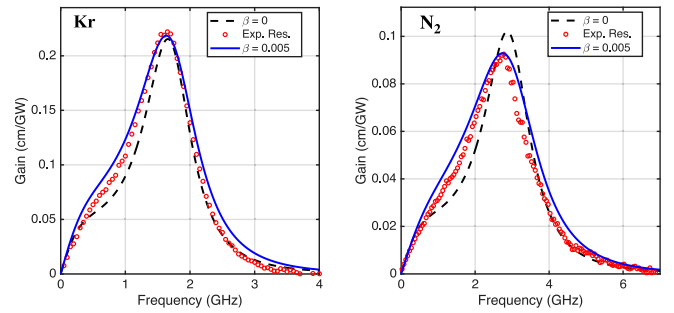


FIG. 6. Comparison of the experimentally measured SBS gain spectra reported in Ref. [26] with the theoretical gain profiles obtained from the conventional kinetic model ($\beta = 0$) and the present velocity-dependent model ($\beta = 0.005$) at a fixed pressure of $P = 6$ atm. The inclusion of velocity-dependent collisional relaxation leads to improved agreement with the experimentally observed peak structure and spectral dependence for Kr and N_2 gases.

conventional BGK description, confirming its consistency as a controlled extension of standard kinetic theory. At the same time, it provides a natural framework for interpreting experimentally observed variations in Brillouin gain spectra that remain unexplained within traditional approaches.

These results identify velocity-resolved relaxation as an additional collisional mechanism contributing to the kinetic description of Brillouin processes. More broadly, the approach developed here enables the systematic incorporation of refined collisional dynamics into models of light-matter interaction and may be extended to other nonlinear scattering phenomena where relaxation effects are significant. In practical terms, this additional degree of freedom offers a pathway for tailoring SBS gain characteristics in high-power laser systems, including applications in laser-driven fusion.

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DATA AVAILABILITY

There are no publicly available research data or software supporting this manuscript. Requests for further information or data should be sent to the authors.

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